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Matemática Básica

EQUAÇÃO DO PRIMEIRO GRAU

Exemplos:

a) $3x - 12 = 0$

$3x = 12 \qquad x = \frac{12}{3} = 4$

b) $3(4x - 3) - (5x - 2) = 3(x - 4)$

$12x - 9 - 5x + 2 = 3x - 12$

$7x - 7 = 3x - 12$

$7x - 3x = -12 + 7$

$4x = -5$

$x = \frac{-5}{4}$

c) $\frac{x}{2} - 6 = x$

$\frac{x}{2} = x + 6$

$x = 2x + 12$

$-12 = 2x - x$

$-12 = x$

$x - 12$

d) $\frac{2(x-1)}{3} + \frac{3x}{2} = \frac{7x-2}{6}$

MMC $3, 2, 6 \mid 2, 3 \} 2 \cdot 3 = 6$

$\frac{4(x-1)}{6} + \frac{9x}{6} = \frac{7x-2}{6}$

$4x - 4 + 9x = 7x - 2$

$4x - 7x + 9x = 4 - 2$

$6x = 2$

$x = \frac{2}{6} = \frac{1}{3}$

e) $5x - \frac{2x-1}{3} + 1 = 3x + \frac{x+2}{2} + 7 \quad \cdot (6)$

$30x - 2(2x - 1) + 6 = 18x + 3(x + 2) + 42$

$30x - 4x + 2 + 6 = 18x + 3x + 6 + 42$

$26x - 21x = 6 + 42 - 8$

$5x = 40$

$x = \frac{40}{5} = 8$

f) $10\left(x + \frac{1}{2}\right) - 6x\left(\frac{1}{x} - \frac{1}{3}\right) = 5$

$10x + 5 - \frac{6x}{x} + \frac{6x}{3} = 5$

$10x + 5 - 6 + 2x = 5$

$12x = 6$

$x = \frac{6}{12} = \frac{1}{2}$